

Chapter 3 Paper 2 — QP + MS Direct Screenshots

Cambridge O Level Physics 5054: Waves and Light up to 3.2.2. This version uses Paper 2 question paper screenshots and the matching Paper 2 mark-scheme page screenshots directly from the updated repository PDFs.

Syllabus area	Covered in sections
3.1 wave motion/features/equation	1, 2, 3, 6, 7
Transverse/longitudinal waves	1, 3, 7
Ripple tank reflection/refraction/diffraction	1, 2, 3, 6, 7
3.2.1 reflection of light / plane mirror	4
3.2.2 refraction, refractive index, critical angle, TIR, optical fibres	3, 4, 5

1) 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank refraction, diffraction — QP screenshot

- 5 (a) Fig. 5.1 is a diagram showing the arrangement of air particles as a longitudinal wave passes through them.

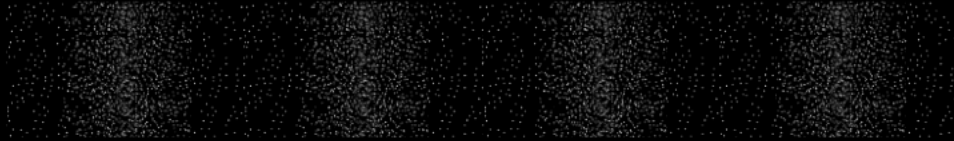


Fig. 5.1

- (i) On Fig. 5.1, mark the centre of a compression with the letter C, and mark the centre of a rarefaction with the letter R. [1]
- (ii) Describe the difference between a compression and a rarefaction.
-
-
- [1]
- (b) In a ripple tank, a water wave is produced by a wooden bar moving up and down on the surface of water.
- (i) The wooden bar makes 45 complete oscillations in 1.0 minute.
- Calculate the frequency of the wave produced.

frequency =Hz [1]

- (ii) The frequency of the water wave is increased by moving the wooden bar up and down more quickly.

State what happens to the speed and what happens to the wavelength of the wave produced.

speed

wavelength

[2]



1) 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank refraction, diffraction — QP screenshot



(iii) The crests of the water wave move into the shallow region shown in Fig. 5.2.

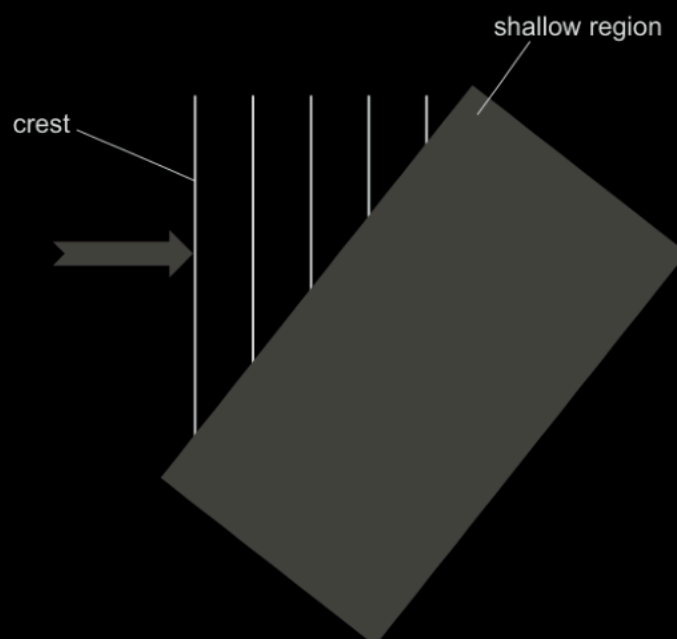


Fig. 5.2

On Fig. 5.2, draw the crests in the shallow region.

[2]

(c) Describe what is meant by the diffraction of a water wave.

.....

.....

..... [1]

[Total: 8]



1) 5054/21/M/J/25 Q5 — longitudinal waves, ripple tank refraction, diffraction — matching MS screenshot

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Question	Answer	Marks
5(a)(i)	C and R marked correctly	B1
5(a)(ii)	compressions are where pressure is high / particles close together / density is high	B1
5(b)(i)	0.75 (Hz)	B1
5(b)(ii)	<i>speed</i> : no change	B1
	<i>wavelength</i> : decreases	B1
5(b)(iii)	at least 3 wavefronts in the shallow region parallel to each other and on correct side of normal	M1
	refracted towards the surface on correct side of surface and join wavefronts shown	A1
5(c)	spreading out / bending at an edge / obstacle or spreading out after passing through a (small) gap / hole	B1

2) 5054/21/O/N/14 Q4 — wave motion, frequency, wavelength, ripple-tank reflection
— QP screenshot

4 Ripple tanks are often used to illustrate wave motion.

(a) Describe what is meant by *wave motion* in a ripple tank.

.....

.....

.....[2]

(b) When describing wave motion, state what is meant by

(i) *frequency*,

.....[1]

(ii) *wavelength*.

.....

.....[2]

(c) A water wave in a ripple tank strikes a barrier. Fig. 4.1 shows some wavefronts of the incident wave.

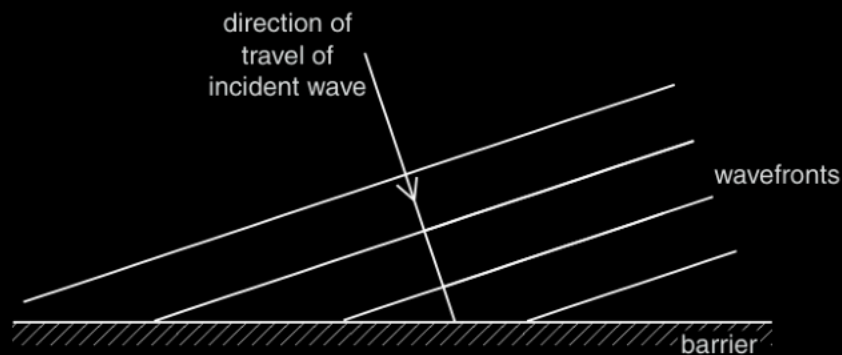


Fig. 4.1

The water wave hits the barrier and is reflected. Three of the wavefronts in Fig. 4.1 have already hit the barrier. The reflected parts of these wavefronts are not shown.

On Fig. 4.1, draw the reflected parts of these three wavefronts.

[3]

**2) 5054/21/O/N/14 Q4 — wave motion, frequency, wavelength, ripple-tank reflection
— matching MS screenshot**

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Section A

- | | | | |
|---|--|--------------------------|-----|
| 1 | <p>(a) (i) $(a =) (v - u) / t$ or $\Delta v / t$ or $(55 - 40) / 2$ or equivalent values from graph
7.5 m/s²</p> <p>(ii) $(F =) ma$ or 180×7.5
1300 / 1350 / 1400 N</p> | C1
A1

C1
A1 | |
| | <p>(b) (i) (acceleration) decreases (to zero)</p> <p>(ii) air resistance / friction / drag mentioned
air resistance / friction / drag increases (with speed) or resultant force decreases (with speed)
(finally) (air) resistance = driving force or resultant is zero</p> | B1

B1
B1
B1 | [8] |
| 2 | <p>(a) (i) $F_1 \times d_1 = F_2 \times d_2$ or $(0.39 \times 0.40) / 0.30$
0.52 N</p> <p>(ii) 0.052 kg or 52 g</p> | C1
A1

B1 | |
| | <p>(b) $(\rho =) m / V$ or $52 / 60$ or $0.052 / 0.000\ 060$ or $0.052 / 60$
870 / 867 / 866.7 kg/m³ or 0.87 g/cm³ or 8.7×10^{-4} kg/cm³ etc.</p> | B1
B1 | [5] |
| 3 | <p>(a) (atoms / molecules / particles) move (about) / collide / hit
(atoms / molecules / particles) collide / hit the walls / surface (of the cylinder)
force on walls (causes pressure)</p> | B1
M1
A1 | |
| | <p>(b) atoms / molecules / particles closer / more compact / more molecules per unit volume / less space to move
more collisions with the wall / surface (of chamber) not if speed / KE changes</p> | B1
B1 | [5] |
| 4 | <p>(a) any two from:
transmission of energy
without net movement of medium
through vibration of particles</p> | B2 | |
| | <p>(b) (i) number of (complete) waves / cycles / oscillations per unit time / second</p> <p>(ii) distance between (neighbouring) waves
distance between (neighbouring) wavefronts / points of same phase or crest to crest / trough to trough distance</p> | B1

C1
A1 | |
| | <p>(c) three reflected wavefronts roughly correct direction
wavelengths equal to each other and incident wavelength by eye
reflected wavefronts joined to incident wavefronts</p> | M1
A1
B1 | [8] |

- 5 Optical fibres are used to transmit telephone signals. Fig. 5.1 shows a ray of light travelling along the inside surface of an optical fibre at P.

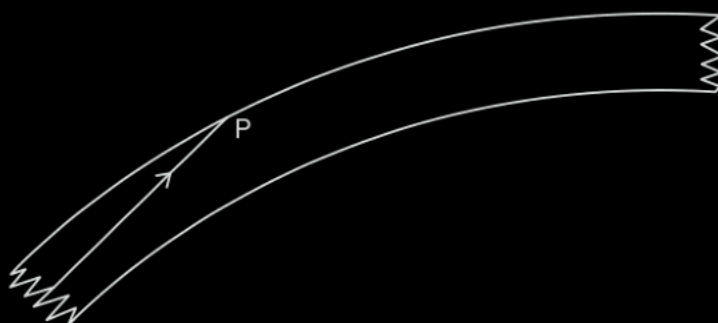


Fig. 5.1

- (a) State one advantage of using optical fibres to transmit telephone signals.

.....
[1]

- (b) (i) On Fig. 5.1, draw a normal at P and mark the angle of incidence with the letter i . [1]

- (ii) State and explain what happens to the ray at P. Use the term *critical angle* in your answer.

.....

[2]

- (c) The optical fibre is made of glass of refractive index 1.5.
 At the start of the optical fibre, the ray enters the glass from air.
 The angle of incidence in the air is 60° .

Calculate the angle of refraction in the glass.

angle =[2]

6 Fig. 6.1 shows a wave on the surface of water. The wave is travelling to the right.

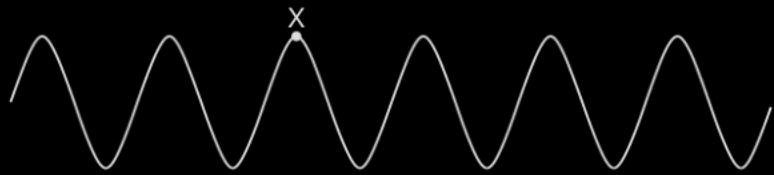


Fig. 6.1 (not to scale)

- (a) Describe what is meant by *wave motion*.
-
-
-[2]
- (b) On Fig. 6.1, draw an arrow to show the direction of the movement of a water molecule at X.
- [1]
- (c) The frequency of the water wave is 2.0 Hz and the wavelength is 2.5 cm.
- (i) Calculate the speed of the wave.

speed =[2]

- (ii) On Fig. 6.1, mark a distance which shows how far a wavefront at X moves in 1.0 s.
- Label this distance *D*.
- [1]

3) 5054/22/M/J/12 Q5-Q6 — optical fibres/TIR + wave motion/ $v=f\lambda$ — matching MS screenshot

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- 4 (a) 80 °C
- (b) $(Q =) mcT$ in any form numerical or algebraic
1530 J C1
A1
- (c) (i) Any 2 lines from
- **latent** heat/energy mentioned
 - **latent** heat/energy given out/lost
 - bonds being made/strengthened
 - molecules lose PE
 - molecules KE constant
- B2
- (ii) ANY 2 lines but max 1 if no change/comparison implied
molecules change **from** OR in liquid
- random arrangement
 - move throughout in some form (e.g. move freely)
 - move or occur in clusters
- change **to** OR in solid
- regular arrangement/shape or fixed position/shape
 - vibrate
 - separation (probably) close(r)
- B2 [7]
- 5 (a) more telephone signals (at one time)
OR great(er) bandwidth; more data (per sec); more signals
OR faster data/information transfer
OR less attenuation; less energy/power/signal loss;
OR long(er) distance (before regeneration)
OR (more) secure
OR less noise/interference OR high(er) quality/clear(er) B1
- (b) (i) correct normal and angle marked B1
- (ii) total internal reflection B1
angle of incidence is larger than critical angle B1
- (c) $(n =) \sin i / \sin r$ in any form numerical or algebraic C1
35(.2644)° **unit ° needed** A1 [6]
- 6 (a) Any 2 of
- an oscillation/vibration/movement up and down
 - carries energy
 - no (net) movement of the medium/transfer of matter
- B2
- (b) arrow downwards or upwards or both B1

3) 5054/22/M/J/12 Q5-Q6 — optical fibres/TIR + wave motion/ $v=f\lambda$ — matching MS screenshot

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(c) (i) ($v = f\lambda$) in any form numerical or algebraic
 $5(.0)\text{ cm/s}$ or $0.05(0)\text{ m/s}$

(ii) line or indication **labelled D** of length 2 wavelengths

B1

7 (a) three lines from one sphere to the other **and** some lines should spread out as they leave one sphere **and** come together nearing the other

B1

correct direction on at least one line and none wrong

B1

(b) ($I = Q/t$) in any form numerical or algebraic
 $2.4 \times 10^{-3}\text{ A}$

C1

A1 [4]

4) 5054/22/O/N/12 Q10 — reflection experiment, optical fibre, plane mirror image — QP screenshot

- 10 A laser produces red light of frequency $4.7 \times 10^{14} \text{ Hz}$. The speed of light in glass is $2.0 \times 10^8 \text{ m/s}$.

(a) Calculate the wavelength in glass of light from this laser.

wavelength = [2]

(b) Describe an experiment to verify *the law of reflection* for light. You may include a diagram in your answer.

..... [5]

(c) Fig. 10.1 shows a ray of light travelling in an optical fibre. The ray strikes the side of the fibre at P.

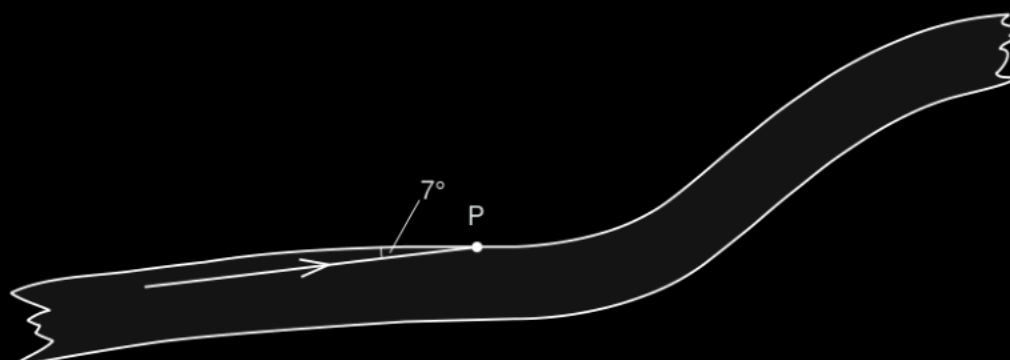


Fig. 10.1

The angle between the ray and the side of the fibre is 7°

4) 5054/22/O/N/12 Q10 — reflection experiment, optical fibre, plane mirror image — QP screenshot

- (i) Determine the angle of incidence of the ray at P.

angle = [1]

- (ii) State and explain what happens to the ray at P.

.....

 [2]

- (d) A room is illuminated by wall lamps. Fig. 10.2 shows a mirror on the wall behind one of the lamps.

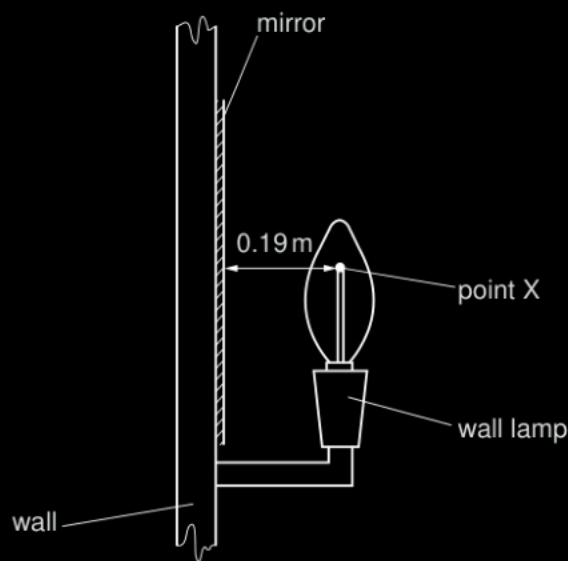


Fig. 10.2 (not to scale)

X is a point on the filament of the lamp. It is 0.19 m in front of the mirror.

- (i) On Fig. 10.2, draw rays from X and locate the image of X. Label the image I. [3]

- (ii) State the distance between I and the mirror.

distance = [1]

- (iii) Suggest one advantage of placing a mirror behind the lamp in the room.

.....
 [1]

4) 5054/22/O/N/12 Q10 — reflection experiment, optical fibre, plane mirror image — matching MS screenshot

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10	(a)	$(\lambda =)v/f$ or $2 \times 10^8/4.7 \times 10^{14}$ $4.3 \times 10^{-7} \text{ m}$			
	(b)	raybox/light source/ laser and mirror shine ray at mirror mark rays measure i and r and equal repeat	pin(s) and mirror place two pins two more pins in line with image measure i and r and equal repeat		
				B1 B1 B1 B1 B1	[5]
	(c)	(i)	83°	B1	
		(ii)	total internal reflection or TIR cao angle of incidence exceeds critical angle	B1 B1	[3]
	(d)	(i)	(at least) one ray from X to mirror (at least) two rays from X to mirror and correct reflections rays traced back to marked I or I marked in correct place (by eye)	M1 A1 B1	
		(ii)	0.19 m	B1	
		(iii)	less/no light wasted or hall brighter	B1	[5]
				[Total: 15]	
11	(a)	(i)	$4.5 + 0.3$ or 4.8 $(I =)V/R$ or 12/4.8 or 12/4.5 or 12/0.3 or 12/0.28125 2.5 A	C1 C1 A1	
		(ii)	decrease resistance (of variable resistor) increase current (in solenoid)	B1 B1	
		(iii)	1. force on PQ/wire or PQ/wire moves force/movement out of page/outwards/towards observer (not upwards) 2. force/speed/acceleration larger	M1 A1 B1	[8]
	(b)	(i)	$(P =)VI$ or 75×12 900 W	C1 A1	
		(ii)	(thick wires) have low resistance (thick wires) not as hot/do not melt	B1 B1	
	(c)		current to relay/coil/solenoid/electromagnet core/relay/coil/solenoid/electromagnet magnetised connections made (in motor circuit)	B1 B1 B1	[7]
				[Total: 15]	

10 A student traces the path of a ray of blue light as it enters and as it leaves a glass prism. Fig. 10.1 shows the trace obtained by the student.

For
Examiner's
Use

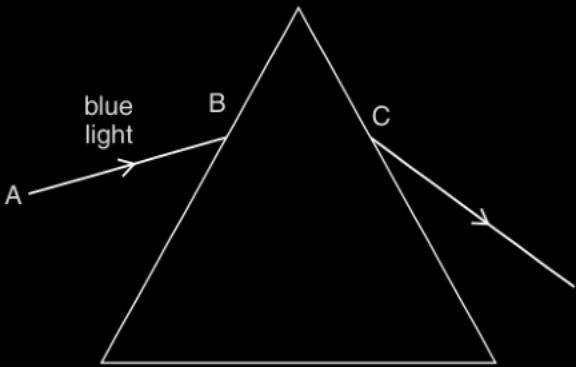


Fig. 10.1

(a) On Fig. 10.1, draw and label, at the point B, the normal, the angle of incidence i and the angle of refraction r . [3]

(b) State, in terms of the properties of light waves, why the light refracts at B.

.....
..... [1]

(c) The angle of incidence for the ray of blue light at B is 45° . The refractive index of the glass is 1.5. Calculate the angle of refraction at B.

angle of refraction = [3]

(d) The student performs another experiment with a ray of red light along the line AB.

On Fig. 10.1, show the path taken by this ray of light as it passes through and leaves the prism. [2]

17

- (e) The student performs another experiment with a semicircular glass block and a ray of white light. Fig. 10.2 shows the path taken by this ray of light as it enters the glass at P until it hits the straight edge at Q.

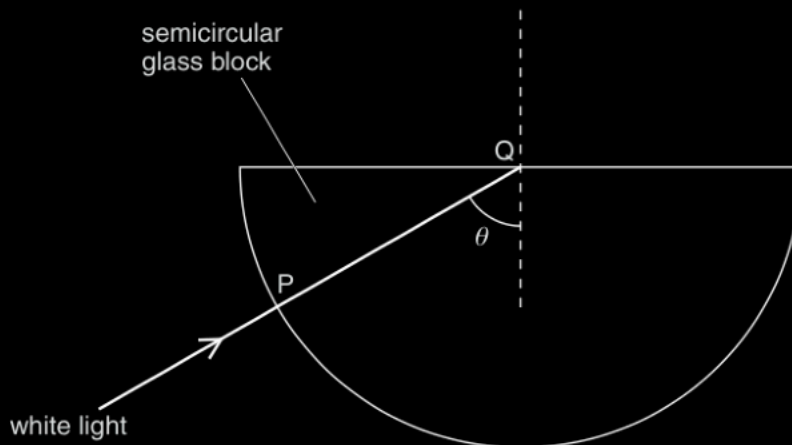
For
Examiner's
Use

Fig. 10.2

The student finds that there is no change in direction as the ray enters the glass at P and that no light passes out of the glass at Q.

- (i) Explain why the ray does not change direction at P.

.....
 [1]

- (ii) Explain why no light passes out of the glass at Q.

.....

 [2]

- (iii) On Fig. 10.2, draw the complete path followed by this ray. [1]

- (iv) The student directs the ray of white light into the glass along different paths, so that the angle θ is slowly reduced.

Describe what happens to the ray at Q.

.....

 [2]

5) 5054/22/M/J/13 Q10 — refraction, Snell's law, critical angle, TIR — matching MS screenshot

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Section B

- 9 (a) (air) molecules hit walls / liquid (surface) B1
- (air) molecules move fast(er) / great(er) kinetic energy B1
- (air) molecules hit more often/more frequently/greater rate / harder / more force B1 [3]
or (liquid) molecules evaporate
- (b) (i) (flask) in (pure) melting ice (and water) B1
- (flask) in (pure) boiling water / above boiling water (at one atmosphere) B1
- (ii) thin(ner) tube
or large(r) flask
or more air/less liquid
or use liquid that expands more (1 mark for each) B2
- (iii) divisions not equally spaced or scale not uniform/not proportional C1
- different distance (along scale) for same temperature rise A1 [6]
or different change in temperature for same distance (along scale)
- (c) (i) ($M=$) $d \times V$ in any form or $1200 \times 5 \times 10^{-5} \times 0.15$ C1
- $9(.0) \times 10^{-3}$ kg; 0.009(0)kg A1
- (ii) 0.09(0) N ecf (i) B1
- (iii) ($P=$) hdg in any form C1
or ($P=$) F/A in any form
- 1800 Pa A1 [5]
- (d) liquids expand less (than air) B1 [1]
or great(er) forces between liquid molecules
- [Total: 15]
- 10 (a) correct normal by eye B1
correct angle of incidence between candidate's normal and incident ray B1
correct angle of refraction marked between candidate's normal and BC B1 [3]
- (b) decrease / change in speed / wavelength B1 [1]

5) 5054/22/M/J/13 Q10 — refraction, Snell's law, critical angle, TIR — matching MS screenshot

Page 6	Mark Scheme	Syllabus	Paper
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- (c) $n = \sin i / \sin r$ seen in any form C1
- ($\sin r =$) $\sin 45^\circ / 1.5$ C1
or 0.47(14) seen
- 28(.1)° C1 [3]

- (d) refracts less at first face and on correct side of normal B1
- refraction at second face away from normal so that red ray and blue ray are diverging B1 [2]

- (e) (i) angle of incidence is 0 B1
or ray along normal/perpendicular to glass
- (ii) angle of incidence/ θ is larger than critical angle B1
total internal reflection occurs B1
- (iii) reflected ray drawn correctly and emerging without refraction from block B1
- (iv) (eventually) light emerges (into air at Q) B1
or light refracts (out at Q)
or (weak) refracted ray appears
- light emerging at Q coloured in some way B1 [6]
or correct description of movement of reflected ray (as θ decreases)

[Total: 15]

- 11 (a) power supply with ammeter and heater in series B1
voltmeter in parallel with heater/ power supply B1 [2]
- (b) (i) ($P =$) VI in any form C1
or 4.2×12
- 50(.4) W A1
- (ii) ($E =$) Pt i.e. any power $\times$ any time e.g. $50(.4) \times 8$ C1
- 8/60 C1
or 0.13(3) seen
or division by 1000 seen anywhere
- 0.0067(2) (kWh) A1 [5]

6) 5054/21/M/J/23 Q7 — diffraction through a gap + wavelength definition — QP
screenshot

- 7 A water wave in a ripple tank diffracts as it passes through the gap in a barrier.

Fig. 7.1 shows a drawing made by a student of the crests in the pattern.

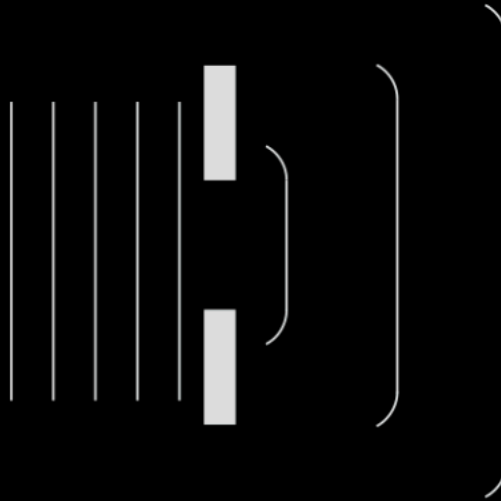


Fig. 7.1

- (a) State **one** way in which the student's drawing of the crests is wrong.

.....
 [1]

- (b) The gap in the barrier is now made smaller than the wavelength, as shown in Fig. 7.2.

Complete Fig. 7.2 with at least three crests to show the new diffraction pattern.



Fig. 7.2

[2]

6) 5054/21/M/J/23 Q7 — diffraction through a gap + wavelength definition — matching MS screenshot

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Question	Answer	Marks
7(a)	wavelength has altered / too large or crests in shadow should be bent / quarter circular above or below gap or vertical lines too straight on two crests	B1
7(b)	at least 3 semicircles centered on the gap	B1
	wavelength constant and the same on right and left of gap and some bending	B1
7(c)(i)	distance between two (consecutive) identical points such as two (consecutive) crests	B1
7(c)(ii)	$v = f\lambda$ numerical or algebraic e.g. $f = 6 / 2$ (Hz)	C1
	3(.0) Hz	A1
7(c)(iii)	1.3 (cm)	B1

7) 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — QP
screenshot

5 Water waves are transverse waves.

(a) Underline **two** other examples of transverse waves.

seismic P-waves

seismic S-waves

sound

X-rays

[1]

(b) Fig. 5.1 shows a wooden bar and a glass block in a ripple tank. The depth of water in the tank is less than the height of the glass block.

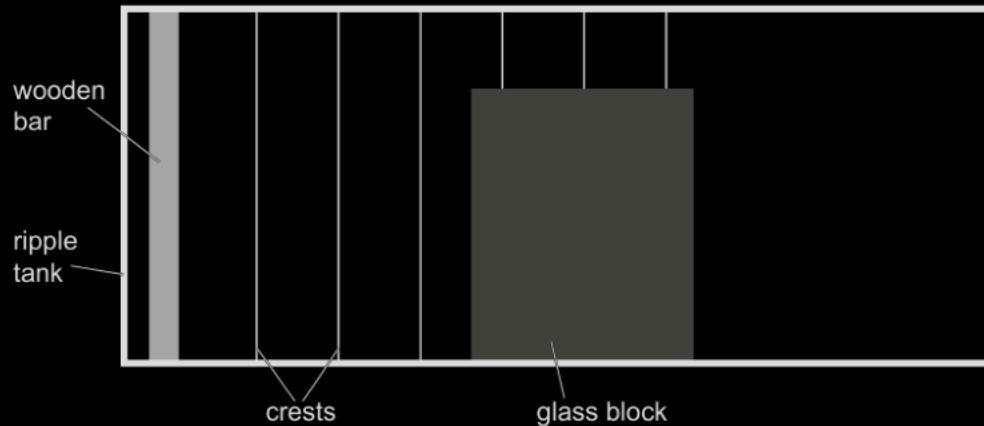


Fig. 5.1 (not to scale)

The wooden bar moves up and down once every 0.15 s to create the crests.

(i) The speed of the water wave is 27 cm/s.

Calculate the frequency and the wavelength of the wave.

frequency = Hz

wavelength = cm
[3]

(ii) The wave diffracts at the right-hand edge of the glass block.

On Fig. 5.1 draw **two** crests after they pass the glass block to show the diffraction. [2]

(iii) Describe how a wave with a smaller wavelength is made with the wooden bar.

.....
..... [1]

(iv) Describe how a decrease in wavelength affects the diffraction.

.....
..... [1]

7) 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — matching MS screenshot

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Question	Answer	Marks
4(a)(i)	–273 (°C) and 0 (K)	B1
4(a)(ii)	particles) move fast(er)	B1
	<u>vibrate</u> / <u>oscillate</u> with larger amplitude	B1
4(a)(iii)	internal energy increases	B1
	temperature stays the same	B1
4(b)(i)	($E = mcT$) in any form numerical or algebraic	C1
	8000 (J)	A1
4(b)(ii)	100–44 or 56	C1
	0.48 or 0.47 (J/(g °C))	A1

Question	Answer	Marks
5(a)	seismic S-waves and X-rays	B1
5(b)(i)	(wavelength =) v / f in any form or (distance =) vt or $f = 1 / T$ or $1 / 0.15$	C1
	frequency 6.7 (Hz)	A1
	wavelength 4.0 (cm)	A1
5(b)(ii)	<u>straight</u> wavefronts above block to the right with same wavelength	B1
	correct diffraction – wavefronts quarter circles with centre at top corner	B1
5(b)(iii)	increase frequency of movement of (wooden) bar	B1

7) 5054/22/M/J/23 Q5 — transverse examples and diffraction at an edge — matching MS screenshot

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Question	Answer	Marks
5(b)(iv)	less diffraction or less bending	B1

Question	Answer	Marks
6(a)(i)	ammeter A ₂ and A ₃ both 0.25 (A)	B1
	voltmeter V ₂ 9(.0 V)	B1
6(a)(ii)	(R =) V / I in any form	C1
	36 (Ω)	A1
6(a)(iii)	current is (directly) proportional to voltage / potential difference	B1
	provided the temperature is constant	B1
6(b)	0.35 (A) seen	B1
	or total resistance $36 + 6 / 0.35$; $36 + 17(.1)$; $53(.1) \Omega$	
	voltage (across R) 0.35×36 or 12.6 (V) seen	B1
	19 V	B1

Question	Answer	Marks
7(a)(i)	X-rays and gamma (radiation / rays)	B1
7(a)(ii)	security marking / detecting counterfeit bank notes / sterilising water / fluorescent effects (e.g. fingerprints, blood, biological fluids) / (sun) tanning / sun beds / cancer treatment / curing inks and resins (e.g. in dental fillings) / production of vitamin (D) / generating electricity / solar panels / disinfection / sterilisation	B1
7(a)(iii)	(skin) cancer / cataracts / burns or destroys the retina or cornea	B1